

XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 5: Dynamics - Applying Newton's Laws of Motion

By:
Dr. Ng Zi Neng
Email: zineng.ng@xmu.edu.my

In this chapter, you'll learn…

- how to use Newton's first law to solve problems involving the forces that act on an object in equilibrium.
- how to use Newton's second law to solve problems involving the forces that act on an accelerating object.
- The nature of the different types of friction forces and how to solve problems that involve these forces.
- How to solve problems involving the forces that act on an object moving along a circular path.
- the key properties of the four fundamental forces of nature.

Newton's Three Laws of Motion

1. A body acted on by no net force, moves with **constant velocity** (which may be zero) and **zero acceleration**.
2. If a net external force acts on a body, the body accelerates. The direction of acceleration is the same as the direction of the net force. **The net force vector is equal to the mass of the body times the acceleration of the body**.
3. If body A exerts a force on body B (an "action"), then body B exerts a force on body A (a "reaction"). These two forces have the same magnitude but are opposite in direction. **These two forces act on different bodies**.

Applying Newton's Laws

- Problem-Solving Strategy When Using Newton's Laws
  - Draw a neat, simple diagram of the system.
  - Isolate each body being analyzed; draw a free-body diagram showing all external forces.
  - Establish convenient axes for each body and find the component forces along these axes; apply Newton's first or second law in component form.
  - Solve the component equations; ensure you have as many independent equations as you have unknowns.
  - Check your results.

Using Newton's First Law When Forces Are in Equilibrium

- An object is in equilibrium when it is at rest or moving with constant velocity in an inertial frame of reference.
- The essential physical principle is Newton's first law:

Newton's first law: *Net* force on an object ... $\sum \vec{F} = 0$ ... must be *zero* for an **object in equilibrium.**

Sum of x-components of force on object must be zero. $\sum F_x = 0$

Sum of y-components of force on object must be zero. $\sum F_y = 0$

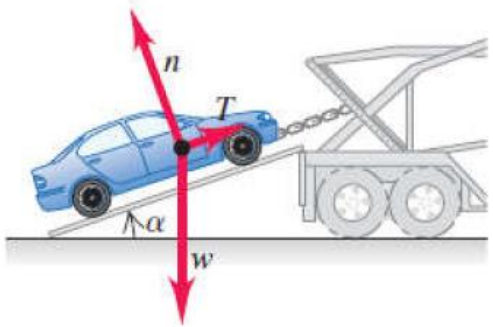
(4.3)

Applying Newton's 1st Law

Example 5.4 An inclined plane

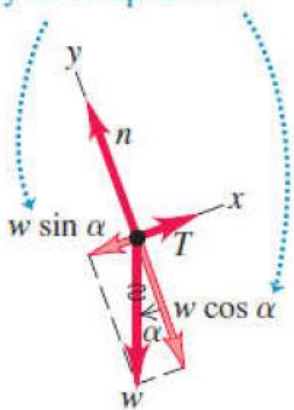
A car of weight w rests on a slanted ramp attached to a trailer (Fig. 5.4a). Only a cable running from the trailer to the car prevents the car from rolling off the ramp. (The car's brakes are off and its transmission is in neutral.) Find the tension in the cable and the force that the ramp exerts on the car's tires.

(a) Car on ramp



(b) Free-body diagram for car

We replace the weight by its components.



Solution

$$\sum F_x = T + (-w \sin \alpha) = 0$$
$$\sum F_y = n + (-w \cos \alpha) = 0$$

$$T = w \sin \alpha$$
$$n = w \cos \alpha$$

Remember that T, w, and n are all *magnitudes* of vectors and are therefore all positive.

Using Newton's Second Law: Dynamics of Particles

- In dynamics problems, we apply Newton's second law to objects on which the net force is not zero.
- These objects are not in equilibrium and hence are accelerating:

Newton's second law: If the *net* force on an object is not zero ...

$$\sum \vec{F} = m\vec{a}$$

... the object has *acceleration* in the same direction as the net force.

Mass of object

(4.6)

Each component of the net force on the object ...

$$\sum F_x = ma_x \qquad \sum F_y = ma_y$$

... equals the object's mass times the corresponding acceleration component.

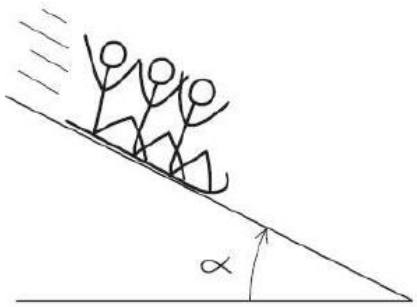
(4.7)

Applying Newton's 2nd Law

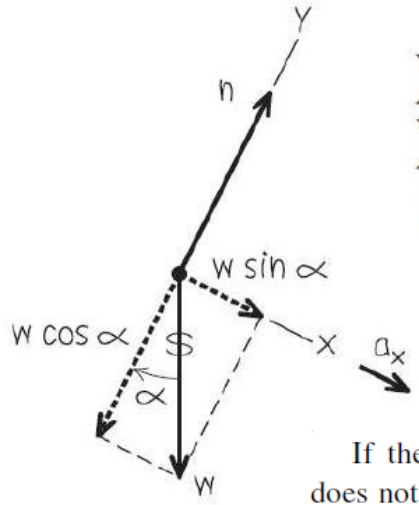
Example 5.10 Acceleration down a hill

A toboggan loaded with students (total weight w) slides down a snow-covered slope. The hill slopes at a constant angle α, and the toboggan is so well waxed that there is virtually no friction. What is its acceleration?

(a) The situation



(b) Free-body diagram for toboggan



$$\sum F_x = w \sin \alpha = ma_x$$

$$\sum F_y = n - w \cos \alpha = ma_y = 0$$

Since $w = mg$,

$$mg \sin \alpha = ma_x,$$

$$a_x = g \sin \alpha$$

If the plane is horizontal, $\alpha = 0$ and $a_x = 0$ (the toboggan does not accelerate); if the plane is vertical, $\alpha = 90°$ and $a_x = g$ (the toboggan is in free fall).

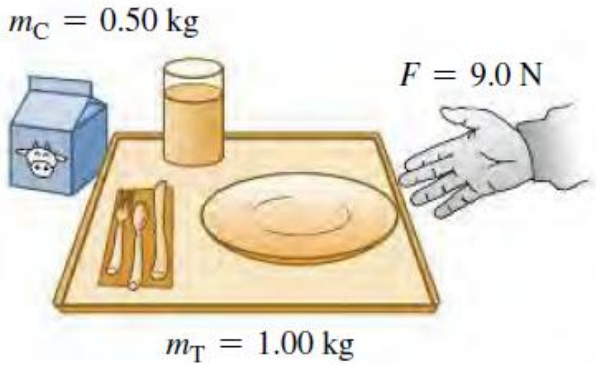
Applying Newton's 2nd Law

Example 5.11 **Two bodies with the same acceleration**

You push a 1.00-kg food tray through the cafeteria line with a constant 9.0-N force. The tray pushes on a 0.50-kg carton of milk (Fig. 5.14a). The tray and carton slide on a horizontal surface so greasy that friction can be neglected. Find the acceleration of the tray and carton and the horizontal force that the tray exerts on the carton.

5.14 Pushing a food tray and milk carton in the cafeteria line.

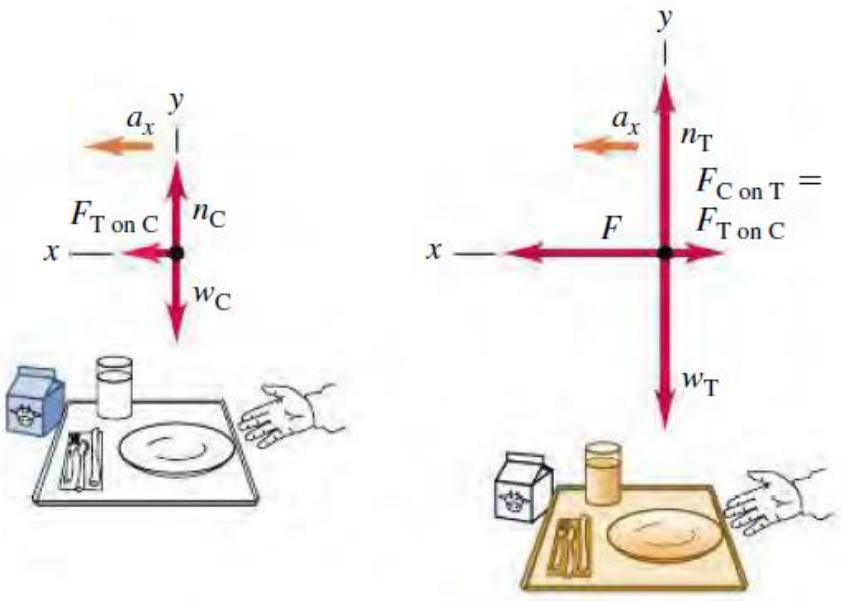
(a) A milk carton and a food tray



Applying Newton's 2nd Law

(b) Free-body diagram for milk carton

(c) Free-body diagram for food tray



Method 1: We treat the carton (mass m_C) and tray (mass m_T) as separate bodies, each with its own free-body diagram. The force F that you exert on the tray doesn't appear in the free-body diagram for the carton.

EXECUTE: *Method 1:* The x-component equations of Newton's second law are

$$\text{Tray:} \quad \Sigma F_x = F - F_{C\text{ on }T} = F - F_{T\text{ on }C} = m_T a_x$$

$$\text{Carton:} \quad \Sigma F_x = F_{T\text{ on }C} = m_C a_x$$

These are two simultaneous equations for the two target variables a_x and $F_{T\text{ on }C}$. (Two equations are all we need, which means that the y-components don't play a role in this example.) An easy way to solve the two equations for a_x is to add them; this eliminates $F_{T\text{ on }C}$, giving

$$F = m_T a_x + m_C a_x = (m_T + m_C) a_x$$

We solve this equation for a_x:

$$a_x = \frac{F}{m_T + m_C} = \frac{9.0\text{ N}}{1.00\text{ kg} + 0.50\text{ kg}} = 6.0\text{ m/s}^2 = 0.61g$$

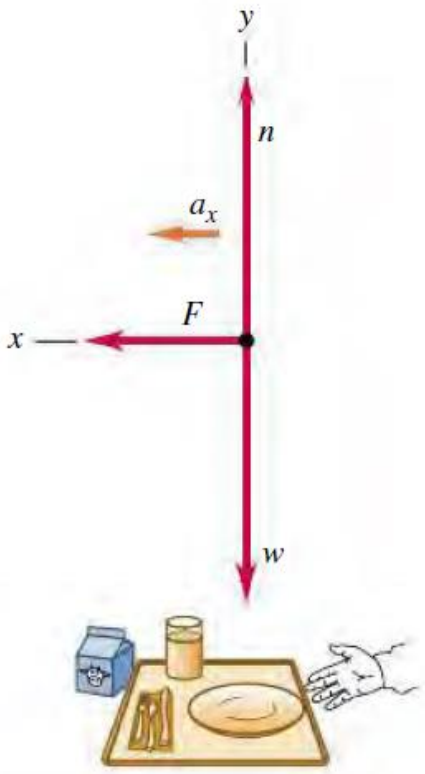
Substituting this value into the carton equation gives

$$F_{T\text{ on }C} = m_C a_x = (0.50\text{ kg})(6.0\text{ m/s}^2) = 3.0\text{ N}$$

Applying Newton's 2nd Law

Method 2: We treat the tray and milk carton as a composite body of mass $m = m_T + m_C = 1.50$ kg. The forces $F_{T \text{ on } C}$ and $F_{C \text{ on } T}$ don't come into play because they're *internal* to this composite body.

(d) Free-body diagram for carton and tray as a composite body



Method 2: The x-component of Newton's second law for the composite body of mass m is

$$\Sigma F_x = F = ma_x$$

The acceleration of this composite body is

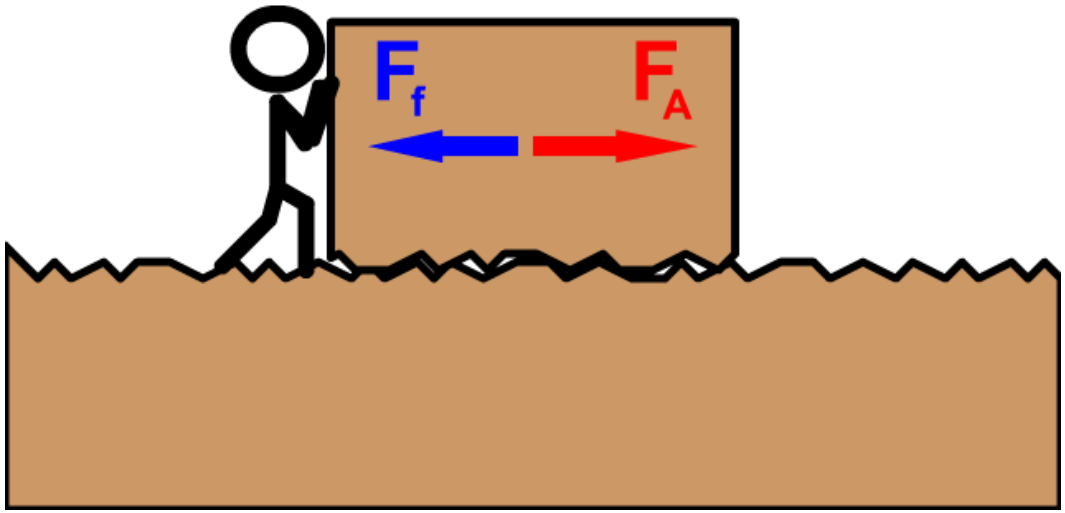
$$a_x = \frac{F}{m} = \frac{9.0 \text{ N}}{1.50 \text{ kg}} = 6.0 \text{ m/s}^2$$

Then, looking at the milk carton by itself, we see that to give it an acceleration of 6.0 m/s^2 requires that the tray exert a force

$$F_{T \text{ on } C} = m_C a_x = (0.50 \text{ kg})(6.0 \text{ m/s}^2) = 3.0 \text{ N}$$

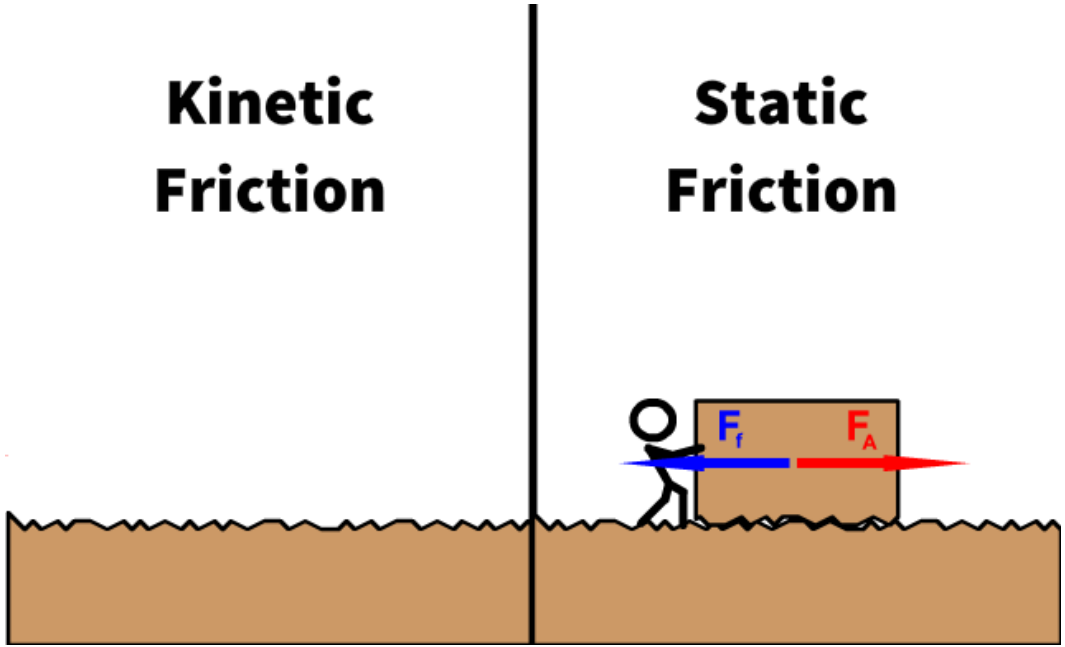
Frictional Forces

- When a body moves on a surface or through a viscous medium (air, water), there are forces of friction because the body interacts with its surroundings.



Frictional Forces

- Force of kinetic friction, $\mathbf{f_k}$: the retarding frictional force on the object in motion.
- Force of static friction, $\mathbf{f_s}$: is the force that counteracts the applied force and keeps the object from moving.



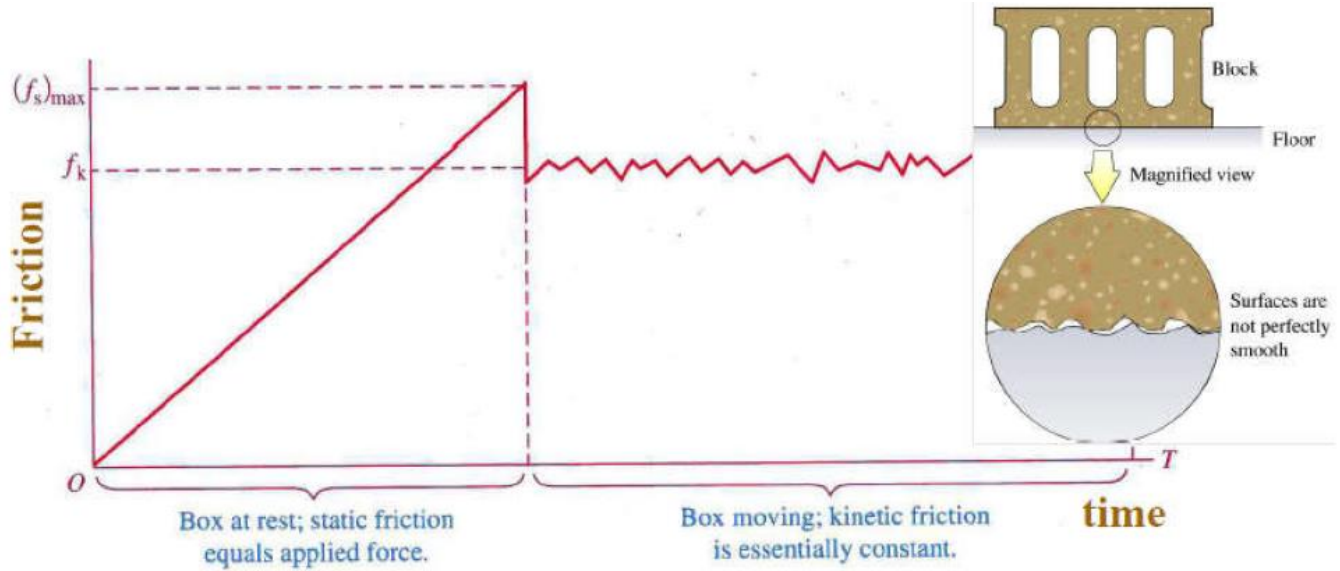
Frictional Forces – Kinetic

- Kinetic friction force, $\mathbf{f_k}$ acts when a body slides over a surface.
- The magnitude of the kinetic friction force f_k usually increases when the normal force increases.
- In many cases the magnitude f_k is found experimentally to be approximately proportional to the magnitude n of the normal force:

Magnitude of kinetic friction force: $f_k = \mu_k n$ (Coefficient of kinetic friction; Magnitude of normal force) (5.3)

- μ_k is the coefficient of kinetic friction.
- The more slippery the surface, the smaller this coefficient. Because it is a quotient of two force magnitudes, μ_k is a pure number without units.

Frictional Forces – Static



- The magnitude of the static friction force, f_s increases as the magnitude of the applied force increases, while keeping the object in place.
- When it is on the verge of moving, f_s is the maximum, $f_{s(max)}$.
- When the applied force exceeds $f_{s(max)}$, the object accelerates and moves.
- Once the object is in motion, the kinetic frictional force f_k becomes less than $f_{s(max)}$.

Frictional Forces – Static

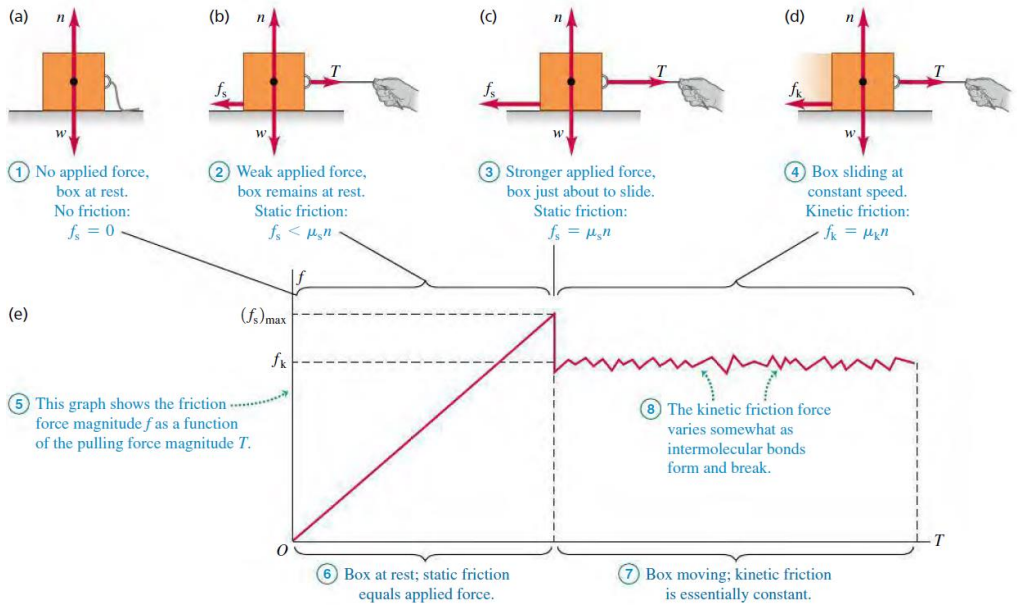
- Experiment shows that in many cases this maximum value, called $f_{s(max)}$, is approximately proportional to n; we call the proportionality factor μ_s the coefficient of static friction.

Magnitude of static friction force — Maximum static friction force

$$f_s \leq (f_s)_{max} = \mu_s n$$

Coefficient of static friction — Magnitude of normal force

(5.4)



Frictional Forces

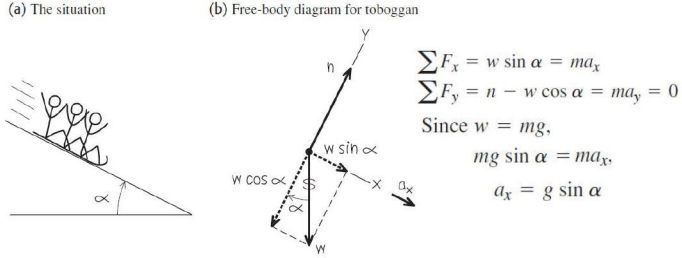
Coefficients of Friction

- The values of μ_s and μ_k depend on the nature of the surfaces (0.05 – 1.5), but μ_k is generally < μ_s.
- The coefficients of friction are nearly independent of the area of contact between the surfaces.
- Although μ_k varies with speed, we normally neglect this.

TABLE 5.1 Approximate Coefficients of Friction

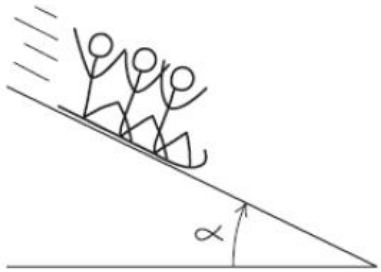
Materials	Coefficient of Static Friction, μ_s	Coefficient of Kinetic Friction, μ_k
Steel on steel	0.74	0.57
Aluminum on steel	0.61	0.47
Copper on steel	0.53	0.36
Brass on steel	0.51	0.44
Zinc on cast iron	0.85	0.21
Copper on cast iron	1.05	0.29
Glass on glass	0.94	0.40
Copper on glass	0.68	0.53
Teflon on Teflon	0.04	0.04
Teflon on steel	0.04	0.04
Rubber on concrete (dry)	1.0	0.8
Rubber on concrete (wet)	0.30	0.25

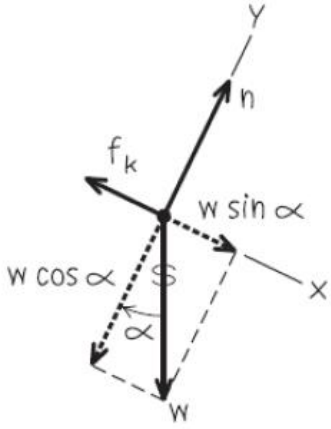
Frictional Forces



EXAMPLE 5.16 TOBOGGAN RIDE WITH FRICTION I

Let's go back to the toboggan we studied in Example 5.10. The wax has worn off, so there is now a nonzero coefficient of kinetic friction μ_k. The slope has just the right angle to make the toboggan slide with constant velocity. Find this angle in terms of w and μ_k.





$$f_k = \mu_k n$$

$$\sum F_x = w \sin \alpha + (-f_k)$$
$$= w \sin \alpha - \mu_k n = 0$$

$$\sum F_y = n + (-w \cos \alpha) = 0$$

Rearranging these two equations, we get

$$\mu_k n = w \sin \alpha \quad \text{and} \quad n = w \cos \alpha$$

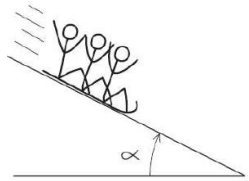
$$\mu_k = \frac{\sin \alpha}{\cos \alpha} = \tan \alpha \quad \text{so} \quad \alpha = \arctan \mu_k$$

Frictional Forces

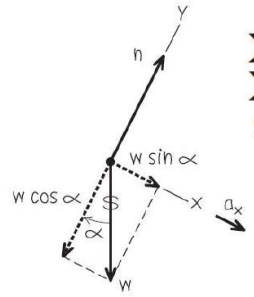
EXAMPLE 5.17 TOBOGGAN RIDE WITH FRICTION II

The same toboggan with the same coefficient of friction as in Example 5.16 *accelerates* down a steeper hill. Derive an expression for the acceleration in terms of g, α, μ_k, and w.

(a) The situation



(b) Free-body diagram for toboggan

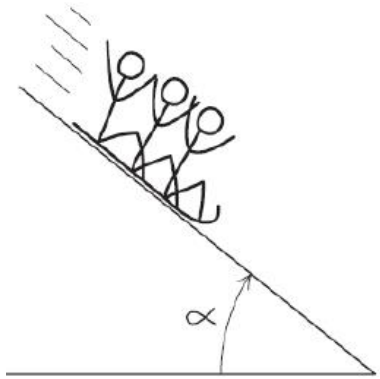


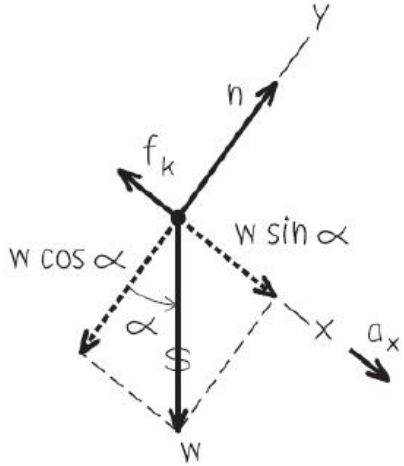
$$\sum F_x = w \sin\alpha = ma_x$$
$$\sum F_y = n - w\cos\alpha = ma_y = 0$$

Since $w = mg$,

$$mg\sin\alpha = ma_x,$$
$$a_x = g\sin\alpha$$

Solution





$$w = mg$$

$$\sum F_x = mg\sin\alpha + (-f_k) = ma_x$$

$$\sum F_y = n + (-mg\cos\alpha) = 0$$

$$n = mg\cos\alpha$$

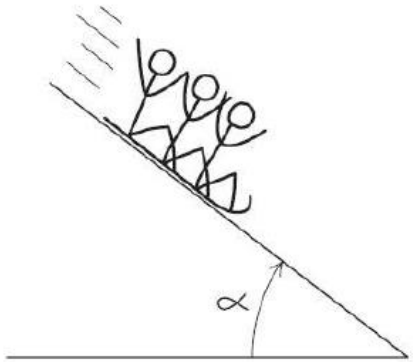
$$f_k = \mu_k n = \mu_k mg\cos\alpha$$

$$mg\sin\alpha + (-\mu_k mg\cos\alpha) = ma_x$$

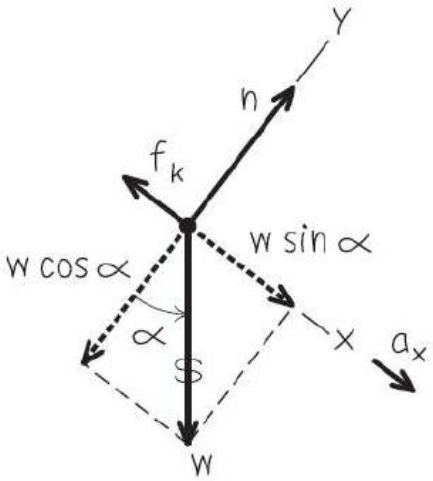
$$a_x = g(\sin\alpha - \mu_k\cos\alpha)$$

Frictional Forces

(a) The situation



(b) Free-body diagram for toboggan



Let's check some special cases. If the hill is vertical ($\alpha = 90°$) so that $\sin\alpha = 1$ and $\cos\alpha = 0$, we have $a_x = g$ (the toboggan falls freely). For a certain value of α the acceleration is zero; this happens if

$$\sin\alpha = \mu_k \cos\alpha \qquad \text{and} \qquad \mu_k = \tan\alpha$$

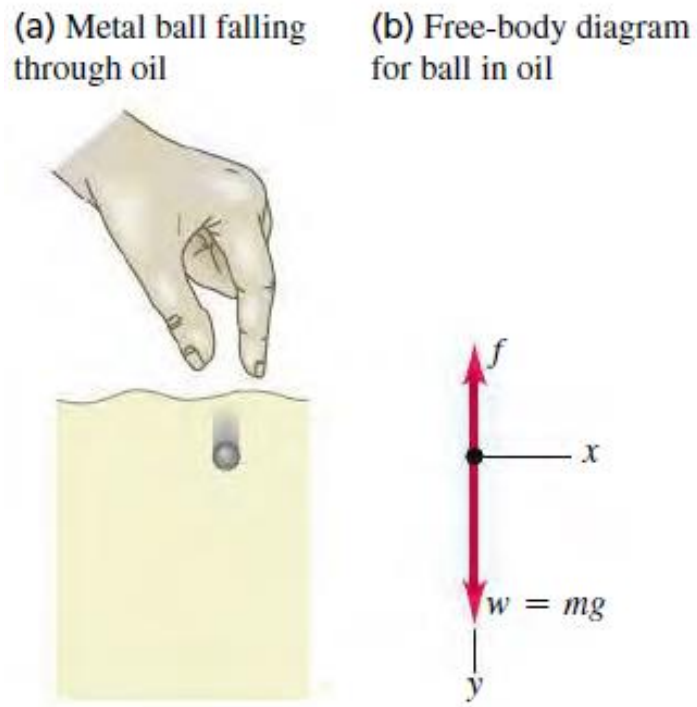
This agrees with our result for the constant-velocity toboggan in Example 5.16. If the angle is even smaller, $\mu_k \cos\alpha$ is greater than $\sin\alpha$ and a_x is *negative;* if we give the toboggan an initial downhill push to start it moving, it will slow down and stop. Finally, if the hill is frictionless so that $\mu_k = 0$, we retrieve the result of Example 5.10: $a_x = g\sin\alpha$.

Notice that we started with a simple problem (Example 5.10) and extended it to more and more general situations. The general result we found in this example includes *all* the previous ones as special cases. Don't memorize this result, but do make sure you understand how we obtained it and what it means.

- It incudes all previous examples(5.10,5.16)

Fluid Resistive Force

- Interaction between moving object and medium cannot be neglected sometimes.
- The medium exerts a resistive force on the object opposite to its direction of motion.
- The fluid resistive force acting on an object depends on the speed of the object.
- Higher the speed, higher the magnitude of resistive force.



Fluid Resistive Force

- The actual dependance is complicated, approximation is therefore used:
  - For small objects at low speeds (e.g. dust in air), where k (unit: N.s/m or kg/s) is a proportionality constant that depends on the shape and size of the body and the properties of the fluid.

$$f = kv \qquad (5.5)$$

  - For large objects at high speeds (e.g. skydiver). Because of the v^2 dependence, air drag increases rapidly with increasing speed and D (unit: $N.s^2/m^2$ or kg/m) depends on the shape and size of the body and on the density of the air.

$$f = Dv^2 \qquad (5.6)$$

Fluid Resistive Force

Resistive Forces at low speed

Neglect upthrust or buoyancy force:

The volume/size of the ball is very small, so its buoyancy can be ignored.

$$\sum F_y = mg + (-kv_y) = ma_y \quad (5.7)$$

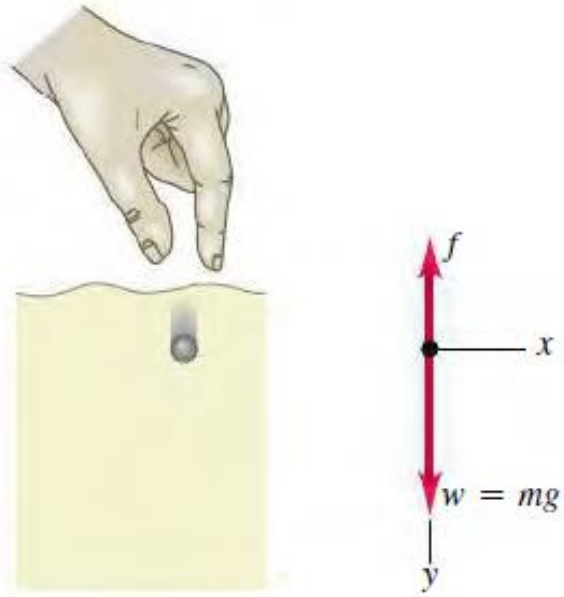
$a_y = dv_y/dt$

$$m\frac{dv_y}{dt} = mg - kv_y$$

$$\frac{dv_y}{dt} = g - \frac{k}{m}v_y$$

(a) Metal ball falling through oil

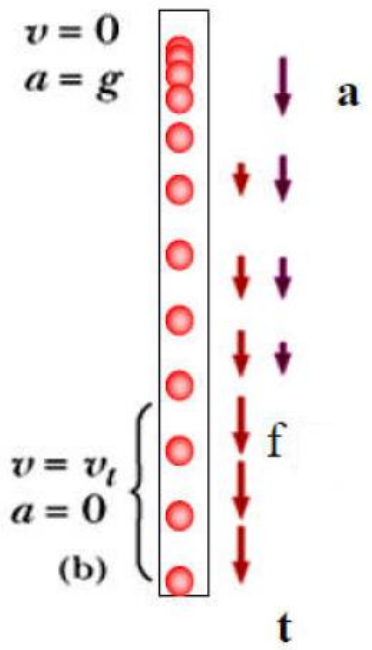
(b) Free-body diagram for ball in oil



Fluid Resistive Force

Terminal speed

$$\frac{dv_y}{dt} = g - \frac{k}{m}v_y$$



- When $v = 0$, the resistive force kv is also zero and the acceleration is simply g.
- As t increases, the resistive force increases and the acceleration decreases.
- Eventually, the acceleration becomes zero and object moves with a constant speed v_t which is the **terminal speed**, given by:

$$mg - kv_t = 0 \quad \Rightarrow \quad v_t = \frac{mg}{k} \qquad (5.8)$$

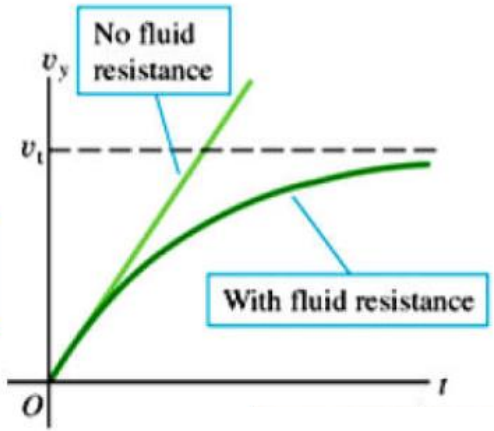
Fluid Resistive Force

Terminal Speed

$$\frac{dv_y}{dt} = g - \frac{k}{m}v_y$$

Solving

$$v_t = \frac{mg}{k}$$



$$v_y = v_t\left(1 - e^{-t/\tau}\right) \quad (5.9)$$

Define (time constant) terminal speed

$$\tau = \frac{m}{k}$$

$$v = \frac{mg}{k}\left(1 - e^{-kt/m}\right)$$

Derivation

Since $\frac{dv}{dt} = g - \frac{k}{m}v$

Rewrite the above equation,

$$\frac{m}{k}\frac{dv}{dt} = \frac{mg}{k} - v$$

Since $v_t = \frac{m}{k}g$, so

$$\frac{m}{k}\frac{dv}{dt} = v_t - v$$

$$\frac{dv}{v - v_t} = -\frac{k}{m}dt$$

$$\int_0^v \frac{dv}{v - v_t} = \int_0^t \left(-\frac{k}{m}\right)dt$$

$$\left(\ln|v - v_t|\right)\Big|_0^v = -\frac{k}{m}t$$

$$\ln\frac{v_t - v}{v_t} = -\frac{k}{m}t$$

$$\ln\left(1 - \frac{v}{v_t}\right) = -\frac{k}{m}t$$

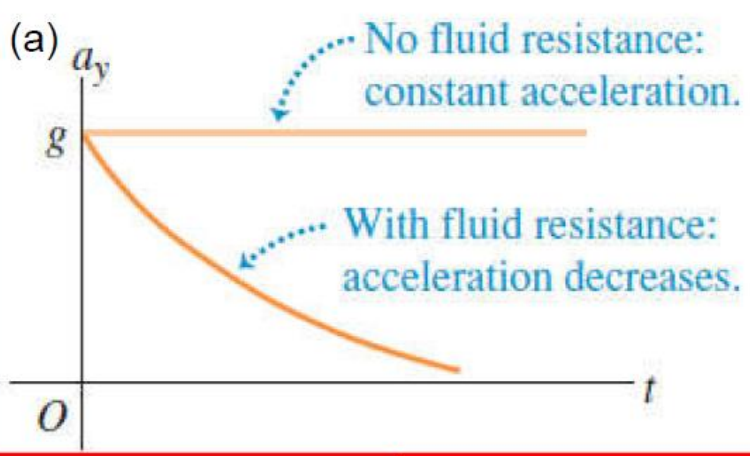
so

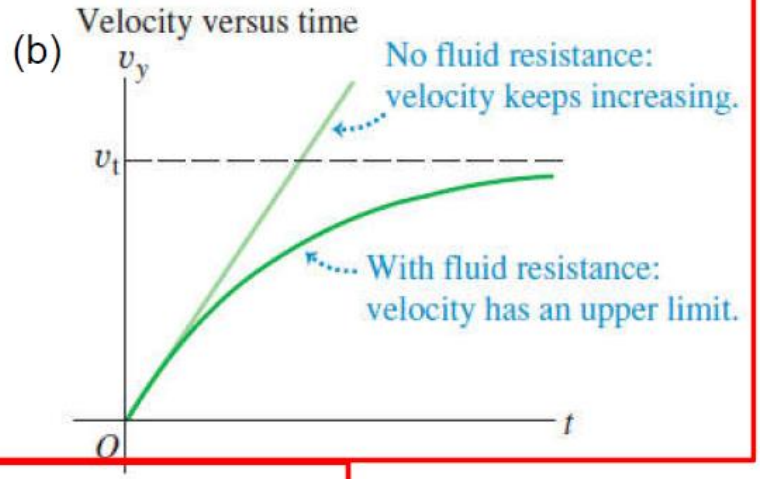
$$1 - \frac{v}{v_t} = e^{-\frac{k}{m}t}$$

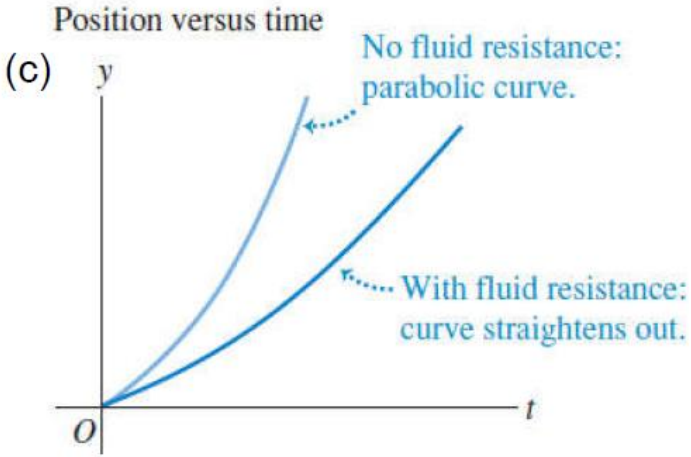
$$v = v_t\left[1 - e^{-\frac{k}{m}t}\right]$$

Fluid Resistive Force

Motion of Falling Body w/ and w/o fluid resistance force

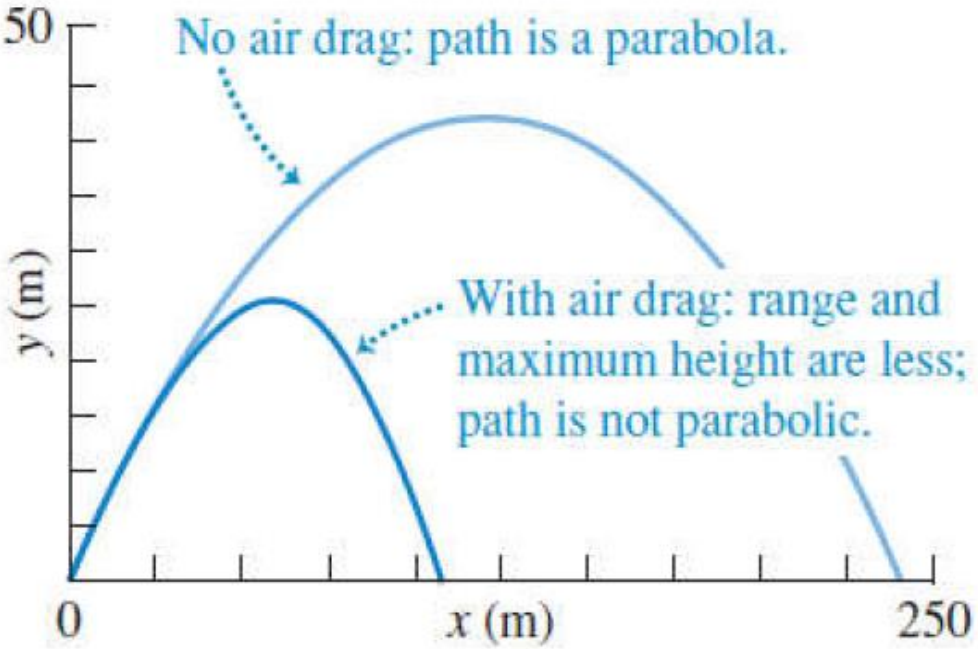






Fluid Resistive Force

- In the Projectile Motion studied in Lecture 3, air resistance is ignored.
- When the air resistance is taken into consideration, both range and maximum height for a projectile will be reduced.
- Air drag is an important part of the game of baseball!



Computer-generated trajectories of a baseball launched at 50 m/s at 35° above the horizontal, assuming coefficient D = 1.3 x 10^{-3} kg/m.

Fluid Resistive Force

Air Drag at High Speeds

- For objects moving at high speeds through air, the resistive force is proportional to the square of the speed Dv^2 in Eq (5.6).
- The terminal speed is reached when Dv^2 equals the weight mg, giving us:

$$f = Dv^2$$ **(Young & Freedman)** (5.6)

$$mg - Dv^2 = ma$$ Equation of motion

$$a = g - \frac{D}{m}v^2$$ Downward acceleration

$a = 0$

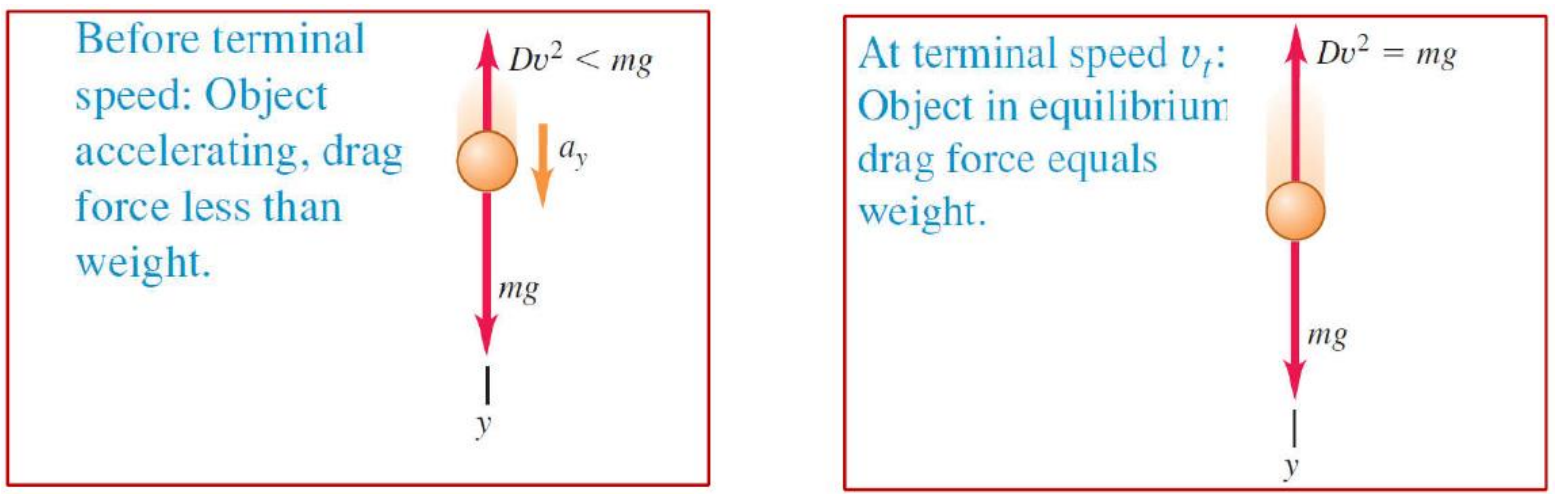
$$v_t = \sqrt{\frac{mg}{D}}$$ Terminal speed

(5.12)

Examples of Fluid Resistive Force

Me and my skydiver falling at terminal speed, in New Zealand. By changing the positions of our arms and legs while falling, we can change the value of the constant *D* in Eq (5.6) and hence adjust the terminal speed of their fall, Eq (5.12).





Air drag and terminal speed.

Examples of Fluid Resistive Force

Example 5.18 Terminal speed of a skydiver



For a human body falling through air in a spread-eagle position (Fig. 5.26b), the numerical value of the constant D in Eq. (5.8) is about 0.25 kg/m. Find the terminal speed for a lightweight 50-kg skydiver.

EXECUTE: We find for $m = 50$ kg:

$$v_t = \sqrt{\frac{mg}{D}} = \sqrt{\frac{(50\ \text{kg})(9.8\ \text{m/s}^2)}{0.25\ \text{kg/m}}}$$

$$= 44\ \text{m/s (about 160 km/h, or 99 mi/h)}$$

Dynamics of Circular Motion

- A particle moving with uniform speed *v* in a circular path of radius *R* experiences an acceleration with magnitude a_{rad} :

Magnitude of acceleration of an object in uniform circular motion

$$a_{rad} = \frac{v^2}{R}$$

Speed of object

Radius of object's circular path

(5.13)

*same as Eq (3.27)

- $\mathbf{a_{rad}}$ or $\mathbf{a_r}$ or $\mathbf{a_c}$ is directed toward the center of the circle, aka **centripetal acceleration**.
- $\mathbf{a_{rad}}$ is always perpendicular to **v**.
- In terms of the period *T*:

$$T = \frac{2\pi R}{v}$$

(5.14)

$$a_{rad} = \frac{4\pi^2 R}{T^2}$$

(5.15)

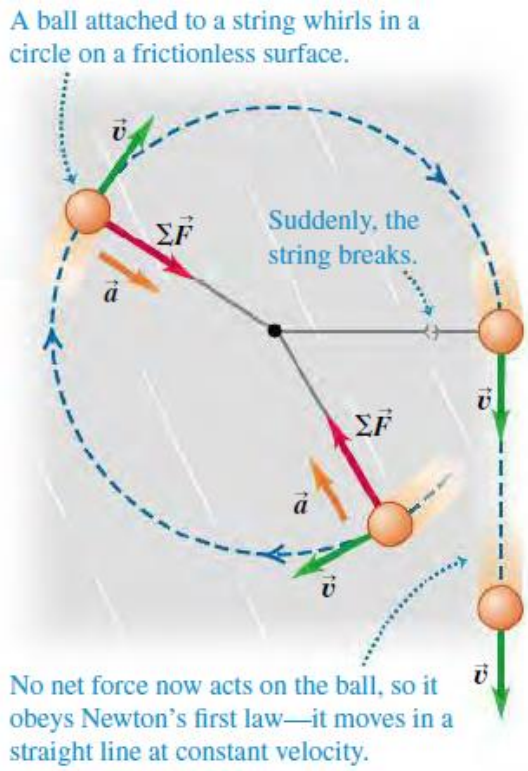
*same as Eq (3.28) and (3.29)

Force of Uniform Circular Motion

- Apply Newton's 2nd law along the radial direction, the magnitude F_{net} of the net force $\sum\vec{F}$ on a particle with mass m in uniform circular motion must be:

$$F_{net} = ma_{rad} = m\frac{v^2}{R} \qquad (5.16)$$

- A force is causing a centripetal acceleration acts toward the center of the circular path and causes a change in the direction of the velocity vector (central force).
- The net force $\sum\vec{F}$ of a uniform circular motion is always directed toward the center of the circle and has a constant magnitude.

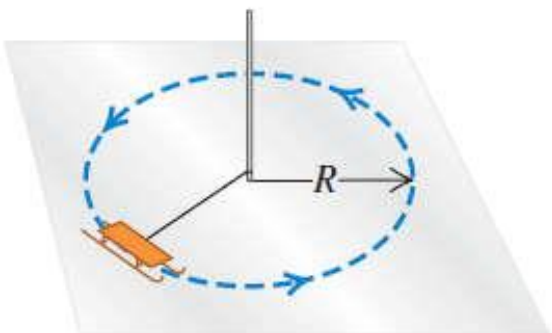


Force of Uniform Circular Motion

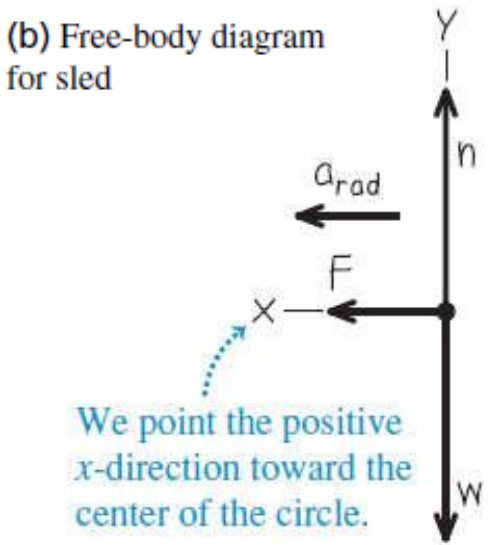
Example 5.19 **Force in uniform circular motion**

A sled with a mass of 25.0 kg rests on a horizontal sheet of essentially frictionless ice. It is attached by a 5.00-m rope to a post set in the ice. Once given a push, the sled revolves uniformly in a circle around the post (Fig. 5.31a). If the sled makes five complete revolutions every minute, find the force F exerted on it by the rope.

(a) A sled in uniform circular motion



(b) Free-body diagram for sled



The force F appears in Newton's second law for the x-direction:

$$\sum F_x = F = ma_{\text{rad}}$$

We can find the centripetal acceleration a_{rad} by using Eq (5.15).

$$T = (60.0\ \text{s})/(5\ \text{rev}) = 12.0\ \text{s}$$

$$a_{\text{rad}} = \frac{4\pi^2 R}{T^2} = \frac{4\pi^2(5.00\ \text{m})}{(12.0\ \text{s})^2} = 1.37\ \text{m/s}^2$$

The magnitude F of the force exerted by the rope is then

$$F = ma_{\text{rad}} = (25.0\ \text{kg})(1.37\ \text{m/s}^2)$$

$$= 34.3\ \text{kg}\cdot\text{m/s}^2 = 34.3\ \text{N}$$

Force of Uniform Circular Motion

Example of uniform circular motion and frictional force

A car makes a circular turn of radius 20m. The coefficient of static friction between the tires and the road is 0.60. What is the maximum speed at which the car can safely make the turn without sliding?

$$F_s = F_c$$

$$\mu_s F_N = \frac{mv^2}{R}$$

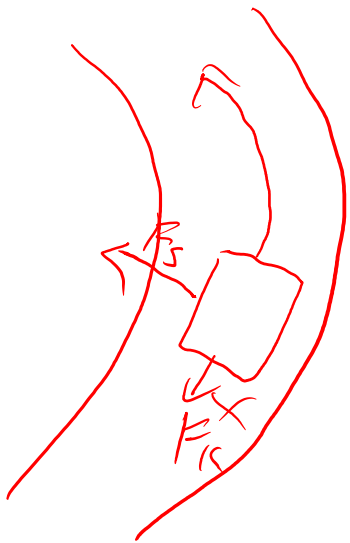
$$\mu_s mg = \frac{mv^2}{R}$$

$$v^2 = \mu_s g R$$

$$v = \sqrt{\mu_s g R}$$

$$= \sqrt{(0.60)(9.8)(20)}$$

$$= 10.84 \text{ m/s}$$



Force of Uniform Circular Motion

Example of uniform circular motion and normal force

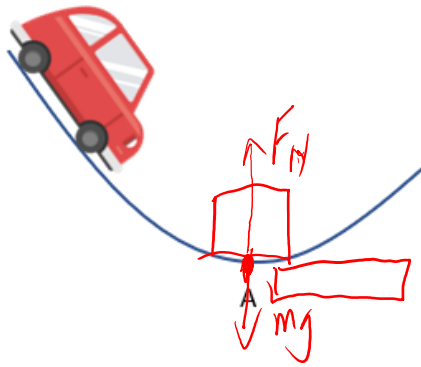
A 1000 kg car travels on a road a speed of 20 m/s. Calculate the normal force exerted by the road at points A and B if the radius of curvature is 100 m. What is the maximum speed the car can travel at point B without losing contact with the road?

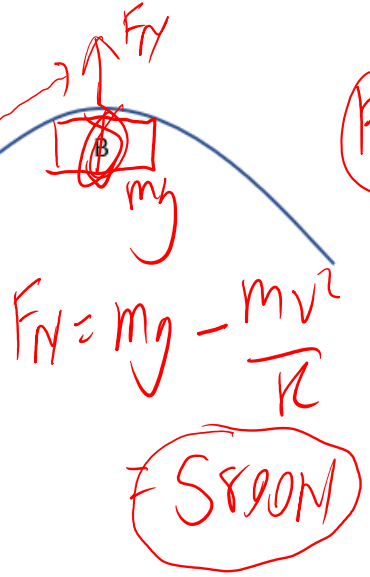
$F_N = \left(mg + \frac{mv^2}{R}\right)$

$= 1000(9.8) + \frac{1000(20)^2}{100}$

$= 13,800\ N$

$a_{rad} = \frac{mv^2}{R}$





$F_N = mg - \frac{mv^2}{R}$

5880N

$F_N = 0$

$mg - \frac{mv^2}{R} = 0$

$v = \sqrt{(9.8)(100}$

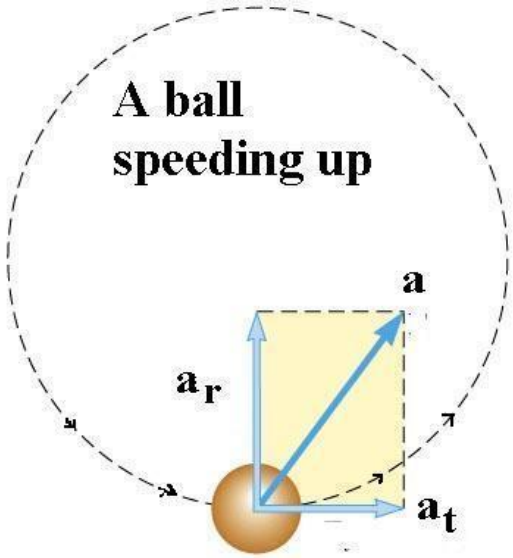
$= 31.3\ m/s$

$mg = \frac{mv^2}{R}$

$v = \sqrt{gR}$

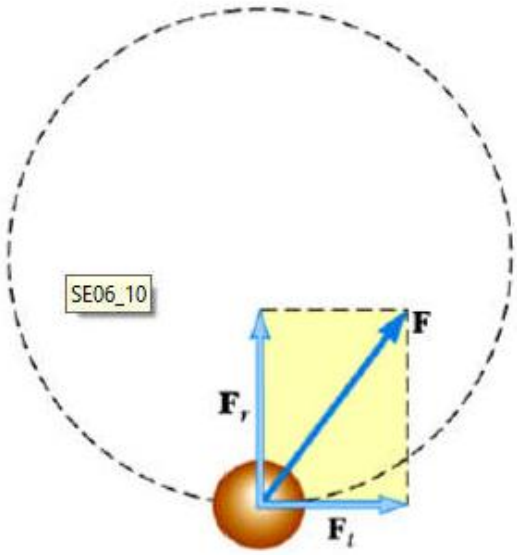
Force of Non-Uniform Circular Motion

- $\mathbf{F}_r$ is directed toward the center of the circle and is responsible for the centripetal acceleration.
- $\mathbf{F}_t$ is tangent to the circle and is responsible for the tangential acceleration (change in speed).



$$\vec{a} = \vec{a}_r + \vec{a}_t$$

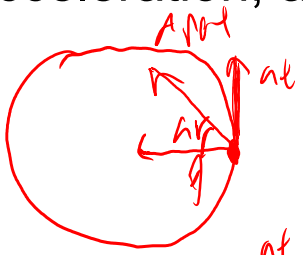
$$\vec{F} = \vec{F}_r + \vec{F}_t$$



Force of Non-Uniform Circular Motion

Example of non-uniform circular motion

A car accelerates from rest at a uniform rate to 40 m/s in 8 seconds on a circular track of radius 600 m. What is the magnitude of the tangential acceleration, radial acceleration, and the net acceleration of the car?



$\tan\theta = \frac{a_t}{a_r}$

$a_{net} = \sqrt{a_r^2 + a_t^2}$

$= \sqrt{2.67^2 + 5^2}$

$= 5.67\ m/s^2$

$v_f = v_0 + at$

$a_t = \frac{v_f - v_0}{t}$

$= \frac{40 - 0}{8}$

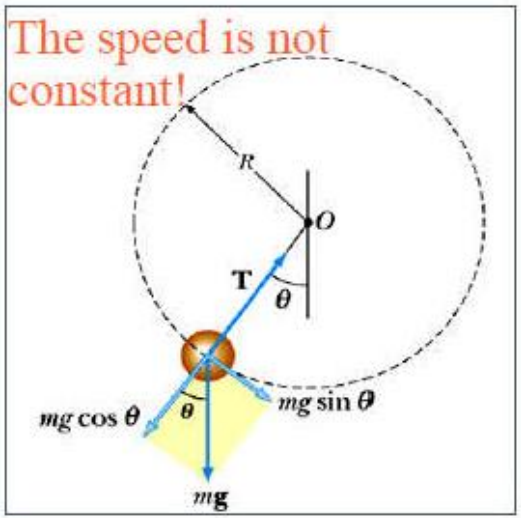
$= 5\ m/s^2$

$a_r = \frac{v^2}{R} = \frac{40^2}{600}$

$= 2.67\ m/s$

Force of Non-Uniform Circular Motion

Mass around a vertical circle



A small sphere of mass m is attached to the end of a cord of length R and whirls in a vertical circle about a fixed point O.
(a)Determine the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.(b) tensions at the top and bottom of the circle

Resolve forces into tangential and radial components:

$$\sum F_t = mg\sin\theta = ma_t$$

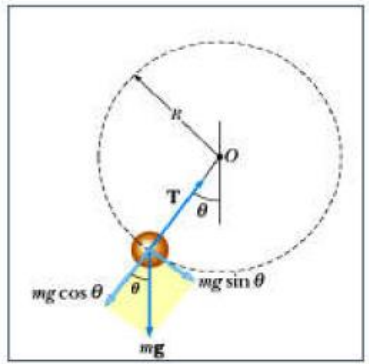
$$\sum F_r = T - mg\cos\theta = \frac{mv^2}{R}$$

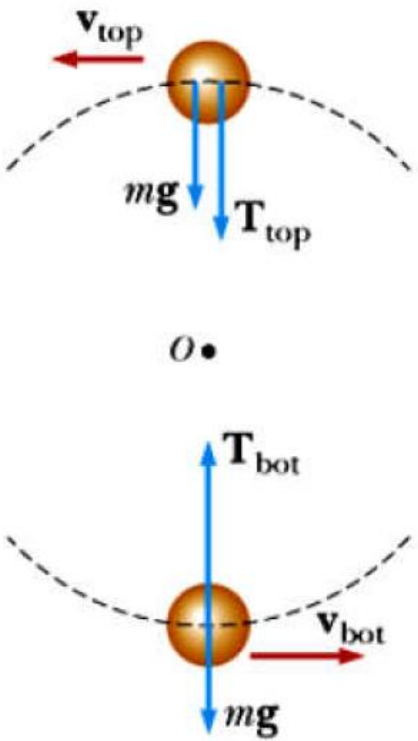
$$T = m\left(\frac{v^2}{R} + g\cos\theta\right)$$

Force of Non-Uniform Circular Motion

Tensions at top and bottom

$$T = m\left(\frac{v^2}{R} + g\cos\theta\right)$$





At the top ($\theta = 180°$)

$$T_{\text{top}} = m\left(\frac{v_{\text{top}}^2}{R} - g\right)$$

Min

At the bottom ($\theta = 0°$)

$$T_{\text{bot}} = m\left(\frac{v_{\text{bot}}^2}{R} + g\right)$$

Max

Force of Non-Uniform Circular Motion

Example of non-uniform circular motion and tension force

A 0.2 kg ball attached to a 1.5 m string moves in a vertical circle. What is the minimum speed that the ball must have at the top so that it will continue to move in a circle?

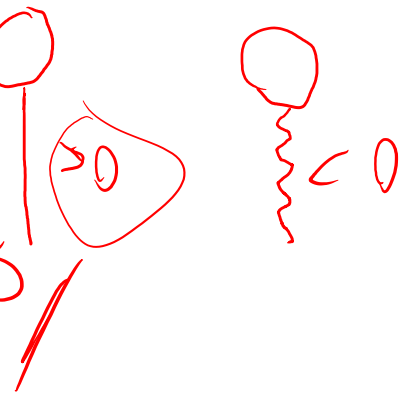
Top: $\Rightarrow T = m\left(\frac{v^2}{R} + g\cos\theta\right) = 0$

$\frac{v^2}{R} = g$

$v = \sqrt{Rg} = \sqrt{1.5(9.8)}$

$= 3.83 \text{ m/s}$





Fundamental Forces

- We have discussed several kinds of forces—including weight, tension, friction, fluid resistance, and the normal force.

How many kinds of forces are there?

- Our **current** understanding is that all forces are expressions of just **four** distinct classes of fundamental forces, or interactions between particles.
  a) Gravitational interaction – force of your weight
  b) Electromagnetic interaction – electric and magnetic forces
  c) Strong interaction – responsible for holding the nucleus of an atom together
  d) Weak interaction – responsible for a common form of radioactivity called beta decay

(c) The strong interaction

The nucleus of a gold atom has 79 protons and 118 neutrons.



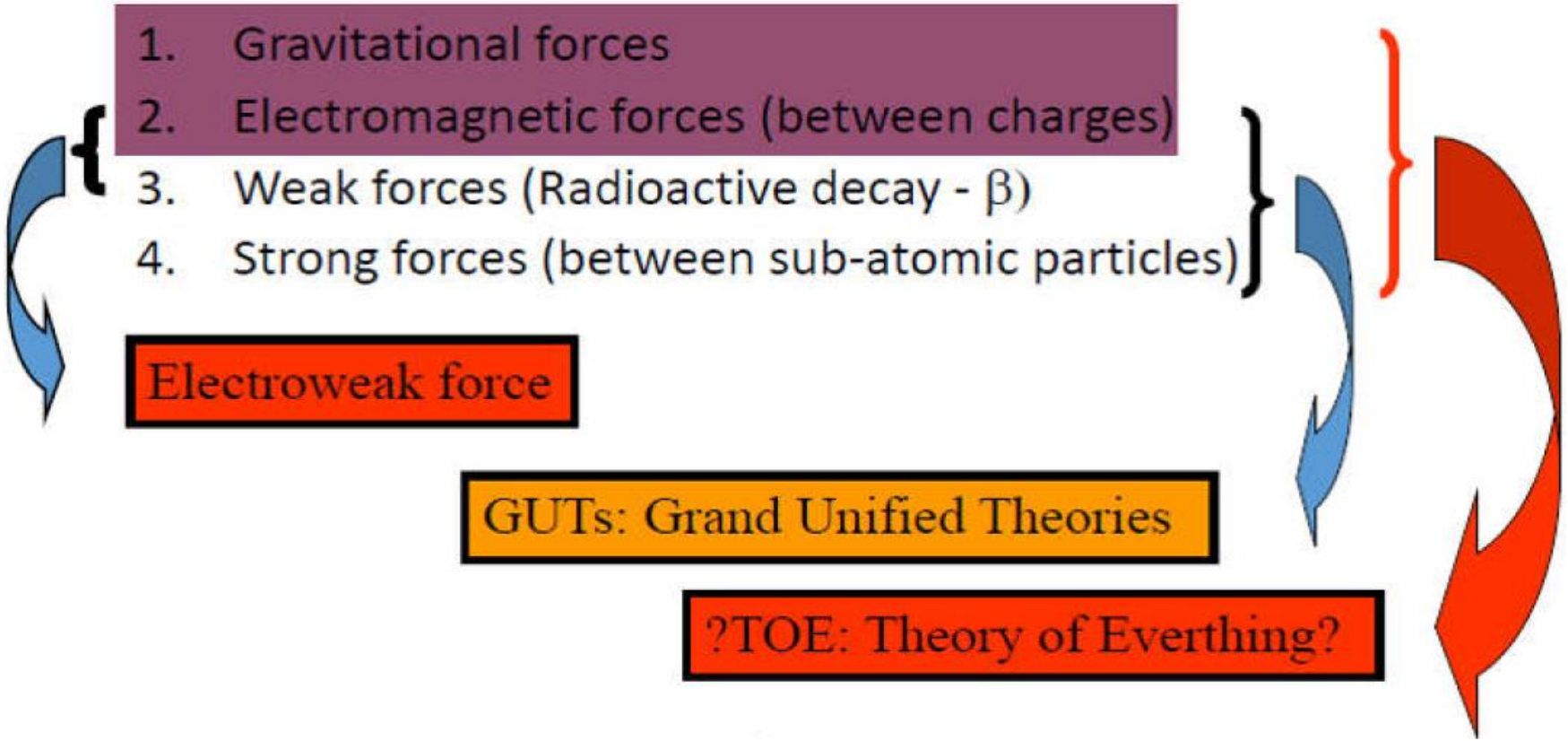
The strong interaction holds the protons and neutrons together and overcomes the electric repulsion of the protons.

(d) The weak interaction

Scientists find the age of this ancient skull by measuring its carbon-14—a form of carbon that is radioactive thanks to the weak interaction.



Fundamental Forces



- Such theories are still speculative, and there are many unanswered questions in this active field of current research.

Summary

- Newton's Laws of Motion are the foundation of Classical Mechanics.
  - Problem-solving when using Newton's Law.
  - Examples on Applying Newton's Law
  - Special topic I: Friction Force + Fluid Resistive Force
  - Special topic II: Dynamics of Circular Motion/Forces of Motion in a Circle
  - Fundamental Forces of Nature
- Textbook
  - Chapter 4 : Newton's Laws of Motion
  - Chapter 5 : Applying Newton's Laws

} **Dynamics**